

AVaaS :

AI Quality Engineering Platform

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Vision Statement: AVaaS is a centralized enterprise AI Quality Hub designed to autonomously test, evaluate, benchmark, and govern AI agents across their lifecycle. Combining zero-integration browser orchestration (Playwright) with Multimodal LLM-as-a-Judge evaluation, AVaaS enables organizations to deploy agentic AI with quantified confidence, security, and ROI.



Industry : QE | Sub-Industry : QA Automation

Title : Agent Validator Platform

Market Limitations vs. The AVaaS Advantage

The Enterprise Testing Gap

- **Legacy UI Automation (Selenium):** Relies on brittle DOM selectors and manual scripting. Cannot evaluate generative reasoning or dynamic text.
- **Backend Evaluators (DeepEval/LangSmith):** Requires heavy SDK instrumentation. Tests backend API payloads but remains completely blind to frontend visual crashes and markdown truncation.

The AVaaS Differentiation

- **Zero-Integration Validation:** Tests the live rendered UI using just the deployment URL. No source code modifications or SDKs required.
- **Multimodal Vision-as-a-Judge:** Captures real client-side screenshots to evaluate visual layout, formatting, and semantic accuracy simultaneously.
- **Autonomous Test Generation:** Instantly auto-generates adversarial and edge-case suites directly from raw agent specifications.

End-to-End Autonomous Validation Pipeline

User/QA Engineer

Web Dashboard (URL)

Phase 1: Autonomous Test Intelligence

Instantly auto-generates normal, edge-case, and adversarial (jailbreak) suites directly from agent specifications.

Phase 2: Resilient Execution Engine

Headless Playwright orchestration with built-in circuit breakers and SPA hydration handling.

Phase 3: Dual-Tier Evaluation:

Tier 1: Deterministic logic and latency constraints.

Tier 2: LLM-as-a-Judge (Safety, hallucination detection, business rules).

Phase 4: Automated Release Gates

Calculates a standardized 0-100 Confidence Score to enforce strict Pass/Fail deployment thresholds.

Web Dashboard/Reports

View Results , Analytics

Expected ROI Indicators

- **Engineering Effort Reduction:** Autonomous prompt-based scenario generation replaces the need for continuous, manual UI test script maintenance.
- **Release Acceleration:** Transitions teams from subjective QA approvals to deterministic, automated Release Gates (e.g., automated PASS/WARN/FAIL status).
- **Defect Leakage Prevention:** Vision-as-a-Judge catches UI-level hallucinations, markdown breaks, and state traps before they impact production users.

Value Realization & Business Impact

Run	Gate Status	Composite	Pass Rate	Risk Score
#287e7477 COMPLETED	WARN	75.0	66%	13.3



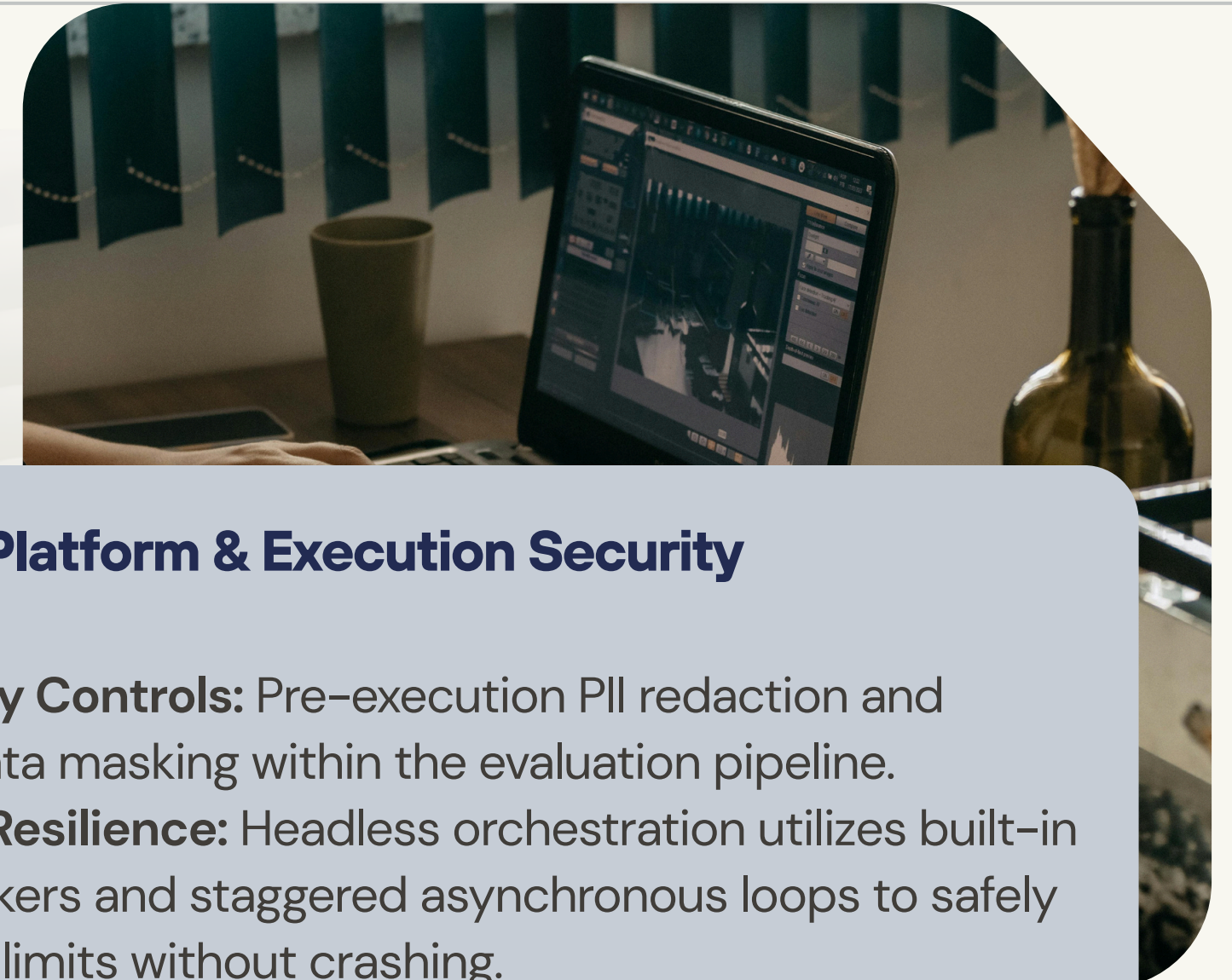
Security, Compliance & Privacy Controls

Responsible AI Governance

- **Adversarial Defense:** Automated injection and jailbreak simulations ensure agents do not leak internal system prompts or bypass safety boundaries.
- **Verifiable Audit Trails:** Every validation run captures full-page visual screenshots, raw text, and token traces to support regulatory compliance.

Platform & Execution Security

- **PII & Privacy Controls:** Pre-execution PII redaction and dynamic data masking within the evaluation pipeline.
- **Execution Resilience:** Headless orchestration utilizes built-in circuit breakers and staggered asynchronous loops to safely handle rate limits without crashing.
- **Secure Deployment:** Dockerized infrastructure designed for secure Azure Virtual Machine hosting directly behind enterprise firewalls.



Scaling to a Centralized AI Quality Hub

Phase 1: Pilot Implementations (Current)

- Onboard initial web-based conversational agents.
- Establish baseline success metrics (Latency, Token Cost, Confidence Score) against existing QA methods.



Phase 2: DevSecOps Integration & Hub Expansion

- Integrate AVaaS directly into enterprise GitHub Actions CI/CD pipelines for automated quality governance.
- Develop reusable, industry-specific testing templates (e.g., Banking, Healthcare compliance accelerators).



Phase 3: Omnichannel & Multi-Agent Orchestration

- Extend testing orchestration beyond web UI to Mobile, API, and voice-based conversational AI.
- Validate complex, end-to-end business processes handled by autonomous multi-agent workflows (CrewAI/LangGraph).

Thank You



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